

PROTOCOL – the shared contract

Status: FROZEN. Changes require the team lead's sign-off and a message in the group, because a change here breaks four other people's work simultaneously.

This document is what lets six people build in parallel. The simulator, the firmware, the gateway, the backend and the frontend all agree on exactly what a message looks like. When real hardware replaces the simulator, **nothing downstream changes.**

1. Identifiers

Field	Format	Example
feeder_id	<UTILITY>- <SUBSTATION>- <FEEDER>	KSEB-TVM-F12
node_id	N- + 3 digits, ascending downstream	N-007

Field	Format	Example
	from the feeder head	
<code>ts</code>	Unix epoch milliseconds , integer, UTC	<code>1757030400123</code>

Node numbering is ordered along the feeder. `N-006` is immediately upstream of `N-007`. This ordering is what makes the fault span computable — do not renumber nodes arbitrarily.

2. Node states

```

NORMAL      E-field within adaptive baseline
             tolerance
SUSPECT     collapse detected locally,
             awaiting neighbour votes
CONFIRMED   quorum reached, fault asserted
RECOVERED   was SUSPECT, field returned
             before quorum (false alarm rejected)
OFFLINE     no telemetry for > 30 s

```

`RECOVERED` matters. It is the state that proves the system rejects false positives, and the frontend must display it prominently.

3. Transport

Environment	Transport
Demo (simulator)	In-process pub/sub inside the backend, then WebSocket to browser
Production	Node → LoRa → Gateway → MQTT over LTE → Backend → WebSocket to browser

MQTT topic scheme:

```
cc/feeder/{feeder_id}/node/{node_id}/telemetry
cc/feeder/{feeder_id}/vote
cc/feeder/{feeder_id}/command
cc/feeder/{feeder_id}/event
```

QoS 1 for telemetry, QoS 2 for `command` and `event` .
Retained flag off for telemetry, on for the last `event` .

4. Message schemas

4.1 `telemetry` — every node, 2 Hz nominal

```
{
  "node_id": "N-007",
  "feeder_id": "KSEB-TVM-F12",
  "ts": 1757030400123,
  "efield_rms": 4.82,
```

```

"baseline": 4.90,
"deviation_pct": -1.6,
"battery_mv": 3720,
"rssi": -94,
"temp_c": 31.4,
"state": "NORMAL",
"seq": 88213
}

```

Field	Type	Unit / range	Notes
<code>efield_rms</code>	float	kV/m, 0.0–20.0	normalised probe output, 10-cycle RMS
<code>baseline</code>	float	kV/m	slow EWMA, τ ≈ 10 min
<code>deviation_pct</code>	float	-100.0 to +100.0	$(\text{efield_rms} - \text{baseline})$ $/ \text{baseline}$ $\times 100$
<code>battery_mv</code>	int	2800– 4200	LiFePO4 terminal voltage
<code>rssi</code>	int	-140 to -30	dBm, last received LoRa packet

Field	Type	Unit / range	Notes
temp_c	float	-10.0 to 60.0	enclosure internal
seq	int	monotonic	replay detection; resets to 0 on reboot

4.2 vote — emitted when a node goes SUSPECT

```
{
  "feeder_id": "KSEB-TVM-F12",
  "asserting_node": "N-007",
  "ts": 1757030400480,
  "votes": [
    { "node_id": "N-006", "agrees": false,
      "deviation_pct": -2.1 },
    { "node_id": "N-008", "agrees": true,
      "deviation_pct": -87.4 }
  ],
  "quorum_required": 2,
  "quorum_met": false
}
```

Quorum is evaluated over the **nearest downstream neighbours**. A break between N-006 and N-007 collapses the field at N-007 and everything

downstream, while N-006 stays normal. That asymmetry is the signature — a global collapse across all nodes is a substation outage, not a break, and must not trip.

4.3 `command` — gateway to actuator

```
{
  "feeder_id": "KSEB-TVM-F12",
  "action": "ISOLATE",
  "reason": "quorum_confirmed",
  "fault_span": ["N-006", "N-007"],
  "ts": 1757030400910,
  "latency_ms": 787,
  "issued_by": "GW-TVM-01",
  "mode": "AUTO"
}
```

`action` ∈ ISOLATE | RESTORE | LOCKOUT | TEST

`reason` ∈ quorum_confirmed | manual |

scheduled_test | watchdog

`mode` ∈ AUTO | ALERT_ONLY — per-feeder

configuration, default ALERT_ONLY

`latency_ms` is measured from the first SUSPECT timestamp to command issue. Surface it in the UI on every event; it is the headline number of the whole project.

4.4 `event` — durable record for the timeline

```
{
  "event_id": "evt_01J9X2K",
  "feeder_id": "KSEB-TVM-F12",
  "type": "CONDUCTOR_BREAK",
  "ts_detected": 1757030400123,
  "ts_confirmed": 1757030400910,
  "fault_span": ["N-006", "N-007"],
  "location": { "lat": 8.5241, "lng":
76.9366 },
  "isolated": true,
  "latency_ms": 787,
  "trail": [
    { "ts": 1757030400123, "node": "N-007",
"state": "SUSPECT" },
    { "ts": 1757030400480, "node": "N-008",
"state": "SUSPECT" },
    { "ts": 1757030400802, "node": "N-009",
"state": "SUSPECT" },
    { "ts": 1757030400910, "node": "GW",
"state": "QUORUM" }
  ],
  "acknowledged_by": null
}
```

```
type ∈ CONDUCTOR_BREAK |
FALSE_POSITIVE_REJECTED | NODE_OFFLINE |
LOW_BATTERY | COMMS_LOSS
```

5. `decide()` — the one shared function

This signature is implemented **once** and reused.
Written in Python in `backend/app/arbiter.py`,
ported to C in `gateway/` and `firmware/`. Do not
write two divergent versions.

```
def decide(
    nodes: list[NodeState],      # ordered
    upstream → downstream
    now_ms: int,
    config: ArbiterConfig,
) -> Decision:
    """
    Pure. No I/O, no clock reads, no
    globals.
    Same input must always give the same
    output – that is what makes it testable
    and what lets the gateway and the cloud
    agree.
    """
```

```
@dataclass(frozen=True)
class ArbiterConfig:
    collapse_threshold_pct: float = -60.0
    # deviation to enter SUSPECT
    sustain_cycles: int = 5
    # consecutive samples required
    quorum_required: int = 2
    # downstream neighbours that must agree
    quorum_window_ms: int = 1500
    # votes older than this expire
    recovery_threshold_pct: float = -20.0
    # deviation to return to NORMAL
```

```

    global_collapse_veto: bool = True
    # all nodes down = outage, never trip

@dataclass(frozen=True)
class Decision:
    action: str          # NONE |
    ALERT | ISOLATE
    reason: str
    fault_span: tuple[str, str] | None
    latency_ms: int | None
    confidence: float     # 0.0-1.0

```

Veto rules — the credibility of the project lives here

1. **Global collapse veto.** If every node on the feeder collapses simultaneously, that is a substation-side outage. Never trip.
2. **Upstream sanity.** The node immediately upstream of the asserted span must still read NORMAL. If it does not, the span is wrong.
3. **Comms loss ≠ fault.** A node going OFFLINE is a maintenance alert, never a vote toward isolation.
4. **Rate limit.** No more than one ISOLATE per feeder per 60 s without manual reset.
5. **Default mode is ALERT_ONLY.** Auto-isolation is opted into per feeder by the utility, not by us.

6. Versioning

Every message carries an implicit schema version via the topic. Breaking changes go to `cc/v2/...`. Until the finale, there is no v2.

7. Test vectors

`docs/vectors/` holds JSON fixtures every implementation must pass:

File	Expected <code>decide()</code> output
<code>break_mid_feeder.json</code>	<code>ISOLATE</code> , span [<code>"N-006"</code> , <code>"N-007"</code>]
<code>break_at_tail.json</code>	<code>ISOLATE</code> , span [<code>"N-010"</code> , <code>"N-011"</code>]
<code>substation_outage.json</code>	<code>NONE</code> (global collapse veto)
<code>rain_burst.json</code>	<code>NONE</code> (recovers before sustain)
<code>vegetation_contact.json</code>	<code>ALERT</code> only, no isolate
<code>single_node_offline.json</code>	<code>ALERT</code> only

File	Expected decide() output
switching_transient.json	NONE

If the Python and C implementations disagree on any vector, the build is broken.